

Supplementary material: Path-spread-resolved quantum transition-state theory for proton transfer

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1 Proof of the harmonic continuum formula

The Euclidean action of a harmonic oscillator $V = \frac{1}{2}m\omega^2 q^2$ is diagonal in Matsubara modes

$$q(\tau) = \frac{1}{\beta\hbar} \sum_{n=-\infty}^{\infty} \tilde{q}_n e^{i\omega_n \tau}, \quad \omega_n = \frac{2\pi n}{\beta\hbar}. \quad (1)$$

Each mode is Gaussian with $\langle |\tilde{q}_n|^2 \rangle = k_B T \beta\hbar / [m(\omega_n^2 + \omega^2)]$. The equal-time correlator is

$$G(0) = \frac{k_B T}{m} \sum_{n=-\infty}^{\infty} \frac{1}{\omega_n^2 + \omega^2} = \frac{\hbar}{2m\omega} \coth\left(\frac{\beta\hbar\omega}{2}\right), \quad (2)$$

where the sum is the standard cotangent partial-fraction identity. The $n = 0$ sector is the centroid, with variance $k_B T / (m\omega^2)$, which is the classical Boltzmann result for a harmonic well.^[1] Subtracting gives

$$\langle \chi_p \rangle = G(0) - \langle \xi_p^2 \rangle, \quad (3)$$

which is Eq. (8) of the main text. Non-negativity follows from $\coth x \geq 1/x$.

Low T : $\coth(\beta\hbar\omega/2) \rightarrow 1$ and the centroid term vanishes, so $\langle \chi_p \rangle \rightarrow \hbar/(2m\omega)$. High T : $\coth x = 1/x + x/3 + O(x^3)$ yields $\langle \chi_p \rangle \rightarrow \hbar^2\beta/(12m) + O(\beta^3)$.

2 Exact finite-bead formula

The P -bead ring polymer is a cyclic Gaussian with eigenvalues

$$\lambda_k = \frac{4mP}{\hbar^2\beta^2} \sin^2(\pi k/P) + \frac{m\omega^2}{P}, \quad k = 0, \dots, P-1. \quad (4)$$

The $k = 0$ eigenvalue is $m\omega^2/P$ and produces the centroid variance $k_B T / (m\omega^2)$. The primitive bead variance minus the centroid variance is

$$\langle \chi_p \rangle_P = \frac{1}{P} \sum_{k=1}^{P-1} \frac{1}{\beta\lambda_k}. \quad (5)$$

As $P \rightarrow \infty$ the sum becomes the Matsubara sum of the previous section.

3 Isotope ordering (harmonic)

With $k = m\omega^2$ fixed, $\omega(m) = \sqrt{k/m}$ and

$$g(m) = \frac{\hbar}{2\sqrt{km}} \coth\left(\frac{\beta\hbar\sqrt{k/m}}{2}\right) - \frac{k_B T}{k}. \quad (6)$$

The second term is mass-independent. The first is $(k_B T/k) u \coth u$ with $u = \beta\hbar\sqrt{k/m}/2$. Since $u \coth u$ is strictly increasing for $u > 0$ and u is strictly decreasing in m , $g(m)$ is strictly decreasing, so $g(m_H) > g(m_D)$. The $T \rightarrow 0$ and $T \rightarrow \infty$ ratios follow by substituting the limits of the previous section. These statements are harmonic; they are not theorems for a general anharmonic surface.

4 Computational parameters

Table 1: Production parameters.

Quantity	Value
Harmonic ω	1000 cm ⁻¹
Harmonic P (H, D)	64, 32
Harmonic production steps	8×10^4
Eckart width a	0.50 Å
Eckart T	300 K
ADW q_0	0.70 Å
ADW α	0.025 kJ mol ⁻¹ Å ⁻¹
ADW umbrellas	11 windows in $[-1, 1]$ Å
ADW k_{umb}	80 kJ mol ⁻¹ Å ⁻²
ADW P (H, D)	32, 24
ADW steps per window	5×10^4
2D k_s	2.0 kJ mol ⁻¹ Å ⁻²
RNG seed	42

Staging proposals use the sequential Lévy construction of the free-particle bridge (end-points of a segment of length $L = P/4$ held fixed; interior beads drawn conditionally). Independent sampling of Brownian-bridge marginals is incorrect and inflates χ_p at large P .

5 Reproducibility

```
python run_all.py --n_steps 80000 --n_equil 15000
python run_all.py --skip_experiments    # figures + compile
```

Dependencies: `numpy`>=1.26, `scipy`>=1.12, `matplotlib`>=3.8. All JSON result files used to generate the figures are in `results/`.

References

- [1] J. Cao and G. A. Voth. The formulation of quantum statistical mechanics based on the Feynman path centroid density. I. Equilibrium properties. *J. Chem. Phys.*, 100:5106–5117, 1994.